

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – I

Roll No.: T016	Name: Himani Malankar
Class: SYIT	Batch:1
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AIM: Getting Started with Tableau and Building Basic Visualizations.

- A. Connect to a CSV dataset and explore the Tableau interface**
- B. Identify and differentiate Measures and Dimensions**
- C. Convert fields between discrete and continuous and observe changes**
- D. Create and customize a basic bar chart**
- E. Build a line chart for time-series analysis**
- F. Create a geographic map visualization**
- G. Combine multiple charts into a simple interactive dashboard**

STEPS:

Step 1: Import the CSV File

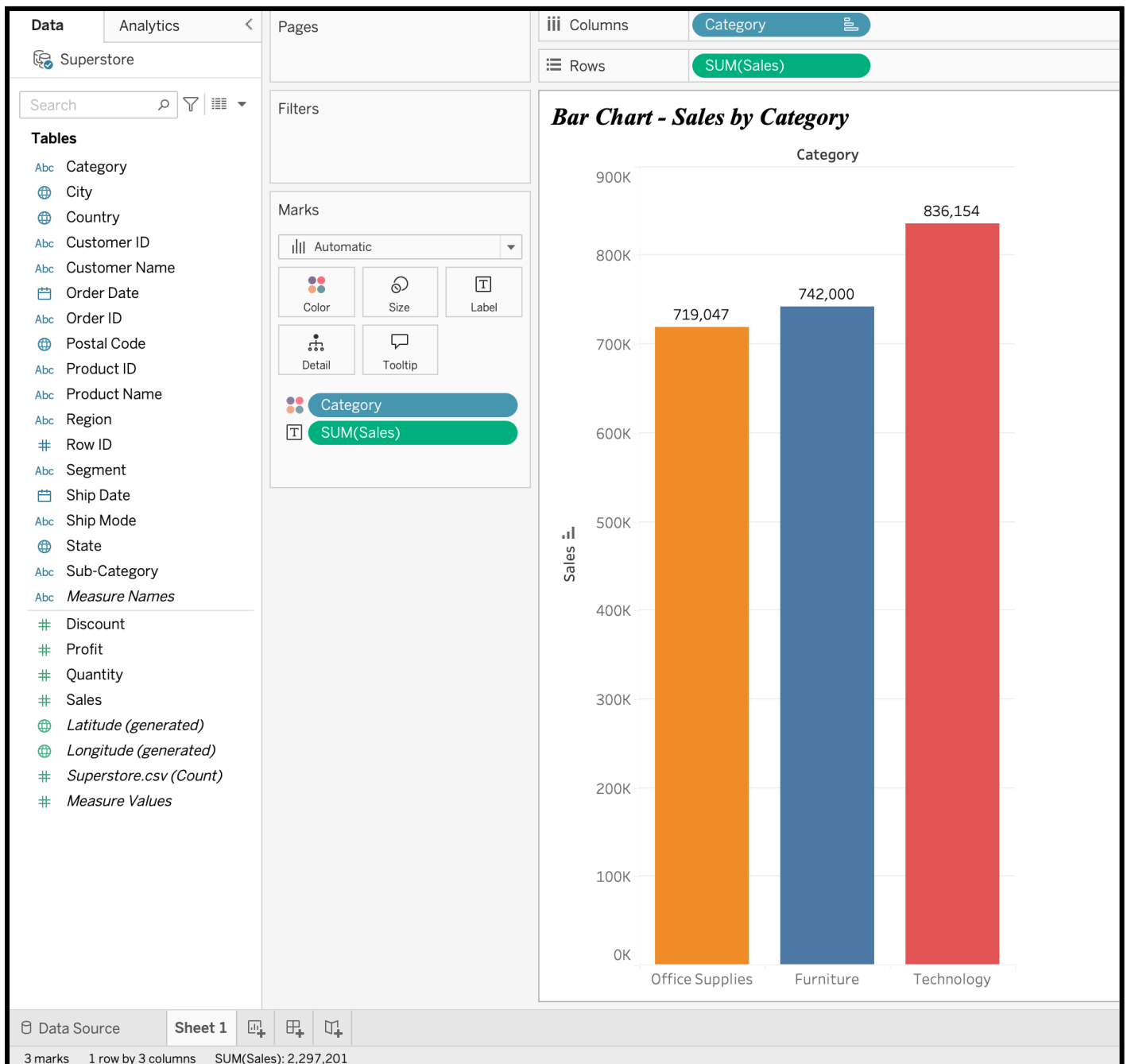
1. Open Tableau Desktop.
2. Click Connect → Text File.
3. Select Superstore.csv.
4. Click Open.
5. Tableau will display the dataset in the Data Source tab.

Step 2: Open a Worksheet

1. Click on Sheet 1 at the bottom.
2. The left panel displays all the fields from the Superstore dataset.
3. Verify that all columns are loaded correctly.

Step 3: Create a Basic Bar Chart

1. Open Sheet 1.
2. Drag Category to the Columns shelf.
3. Drag Sales to the Rows shelf.
4. Tableau automatically creates a Bar Chart.
5. Drag Category to Color to display different colors for each category.
6. Drag Sales to Label to display the sales values on each bar.
7. The chart shows the total sales for each product category.

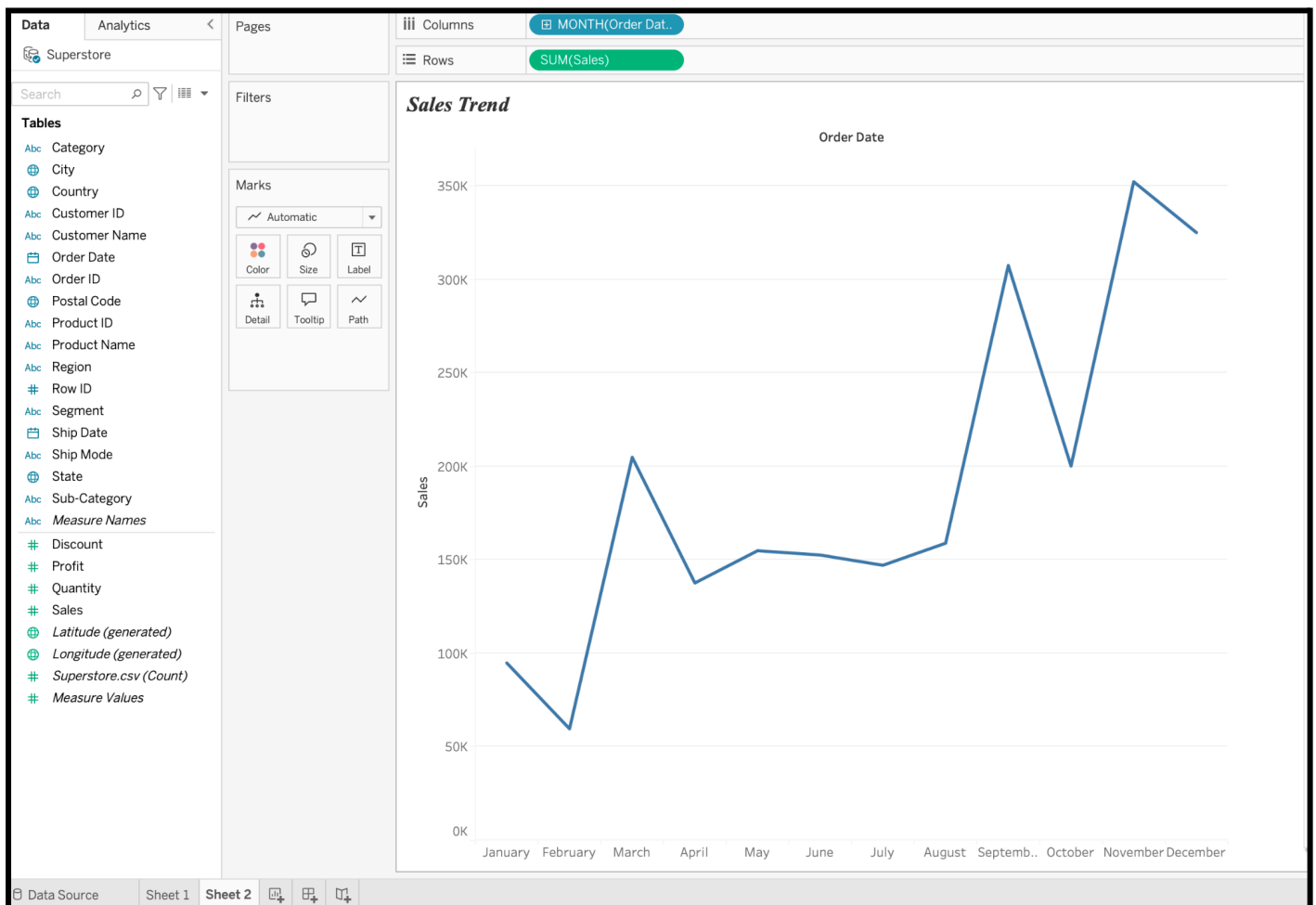


INSIGHT:

- Technology is the best-performing product category.
- Office Supplies contribute the least sales.
- The company can continue investing in Technology products while exploring strategies to improve Office Supplies sales.

Step 4: Create a Line Chart

1. Create a new worksheet and rename it Sales Trend by Month (Line Chart).
2. Drag Order Date to the Columns shelf.
3. Change the date level to Month.
4. Drag Sales to the Rows shelf.
5. Select Line from the Marks card (or use Show Me → Line Chart).
6. The line chart displays the monthly sales trend of the Superstore dataset.

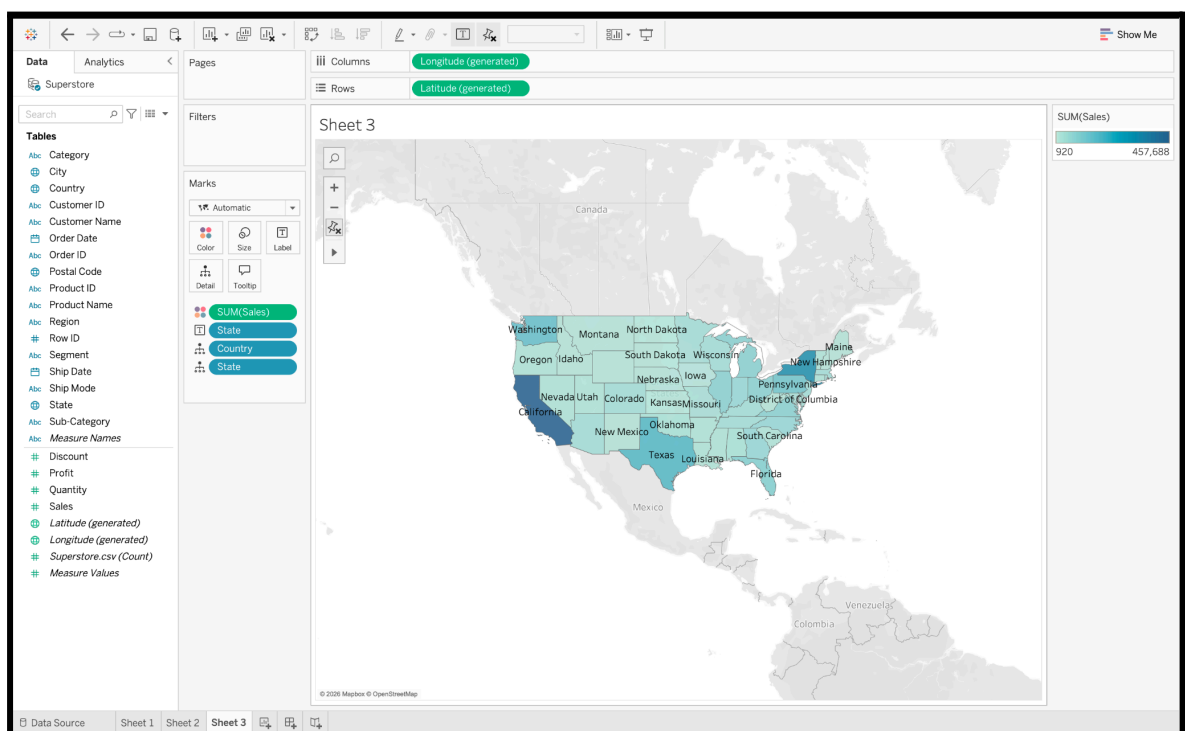
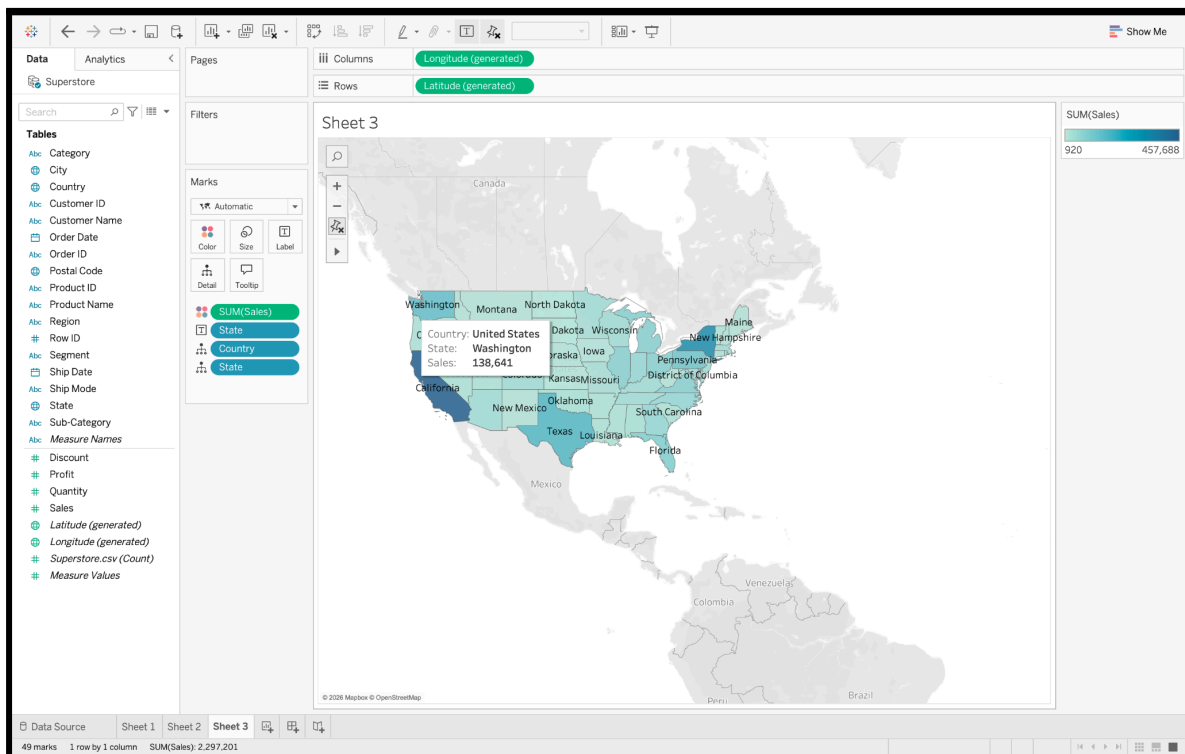


INSIGHT:

- Sales fluctuate throughout the year rather than remaining constant.
- November records the highest sales (around 352K).
- February has the lowest sales (around 60K).
- Sales increased significantly during the last quarter (September to December), indicating a strong year-end sales period.

Step 5: Create a Geographic Map

1. Create a new worksheet and rename it **Sales by State (Geographic Map)**.
2. Double-click the **State** field to generate a map.
3. If the map is not generated correctly, set the Geographic Role of **State** to **State/Province**.
4. **If the map still does not display correctly (dots do not appear), drag the **Country** field from the Data pane and drop it on the **State** field or on **Detail** in the Marks card.** This helps Tableau correctly identify the geographic locations.
5. Drag **Sales** to **Color**.
6. Drag **State** to **Label** or **Tooltip**.
7. The geographic map displays the sales distribution across different states.



INSIGHT:

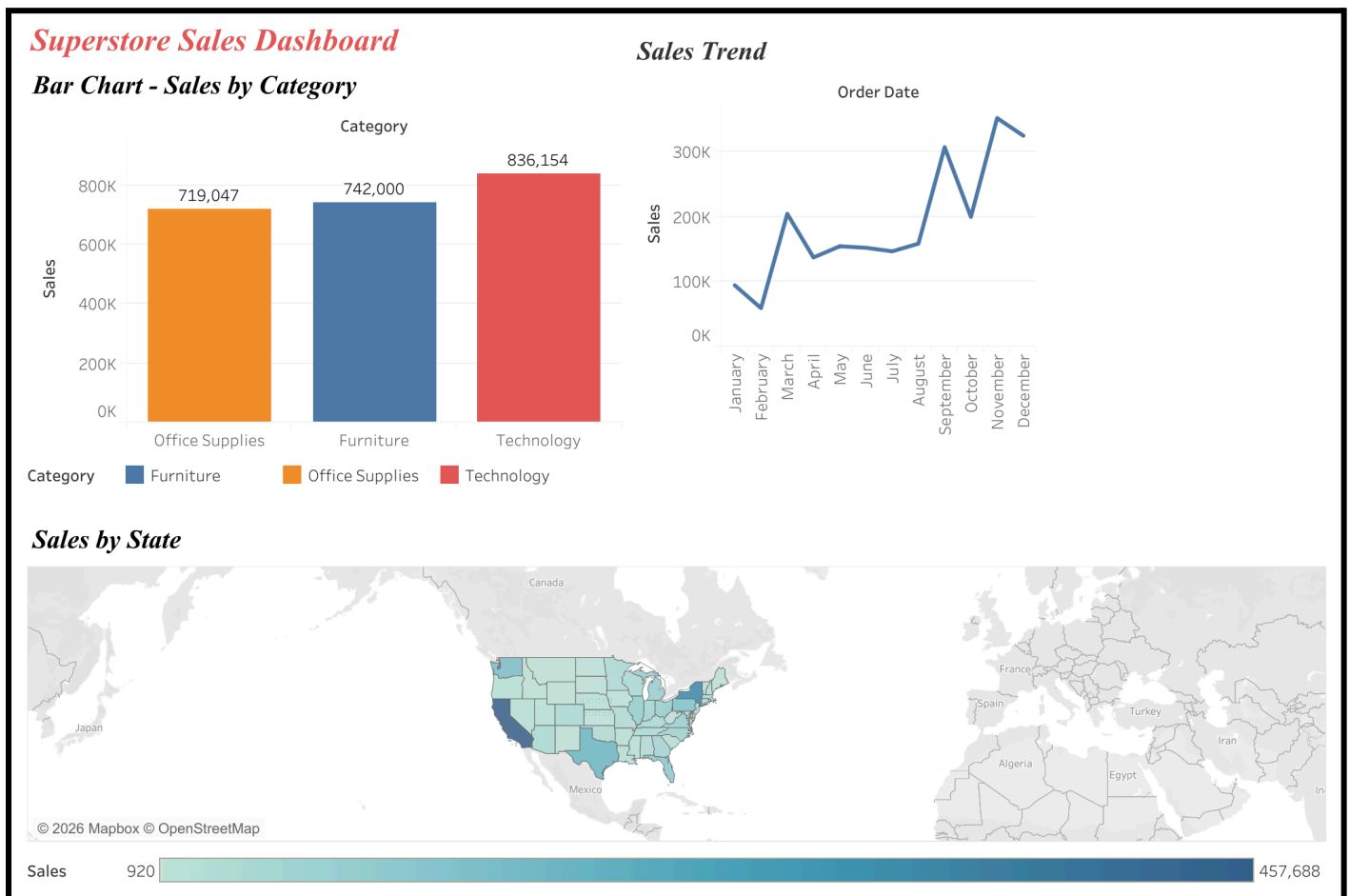
- California has the highest sales, indicated by the darkest shade on the map.
- Several central and western states show moderate sales.
- Some states have lighter shades, indicating comparatively lower sales.
- Sales distribution varies significantly across different states.

Step 6: Create an Interactive Dashboard

Dashboard Name: Superstore Sales Dashboard

1. Click the New Dashboard icon at the bottom of Tableau.

2. Rename the dashboard as Superstore Sales Dashboard.
3. Set the dashboard size to Automatic.
4. Drag the following worksheets from the Sheets pane to the dashboard:
 - Sales by Category (Bar Chart)
 - Sales Trend by Month (Line Chart)
 - Sales by State (Geographic Map)
5. Arrange all the worksheets neatly using the Tiled layout.
6. Resize each visualization so that all charts are clearly visible.
7. Edit the titles of each worksheet:
 - Bar Chart – Sales by Category
 - Sales Trend
 - Sales by State
8. Verify that all visualizations display the data correctly.
9. Save the dashboard.



DASHBOARD INSIGHT:

- Technology is the highest-selling product category with approximately 836K in sales.
- Office Supplies has the lowest sales among the three categories.
- Monthly sales fluctuate throughout the year, with the highest sales recorded in November.
- February records the lowest monthly sales.
- California contributes the highest sales among all states, while many other states contribute moderate sales.
- The dashboard provides a consolidated view of category-wise, time-wise, and state-wise sales performance, making it easier to identify business trends and high-performing regions.

The dashboard shows that Technology is the best-performing product category, sales peak during November, and California records the highest state-wise sales. Combining multiple visualizations in one dashboard enables quick analysis of sales performance across categories, time, and regions.